JAMA Psychiatry | Original Investigation

Age at First Attention-Deficit/Hyperactivity Disorder Diagnosis and Educational Outcomes

Lotta Volotinen, MSc; Hanna Remes, PhD; Pekka Martikainen, PhD; Niina Metsä-Simola, PhD

IMPORTANCE Early diagnosis of attention-deficit/hyperactivity disorder (ADHD) is often recommended, but it is unknown whether age at diagnosis is associated with educational outcomes.

OBJECTIVE To estimate whether age at ADHD diagnosis is associated with school performance, completed degrees, educational enrollment, and school dropout.

DESIGN, SETTING, AND PARTICIPANTS This population-based cohort study used national registry data for individuals born in Finland between January 1, 1990, and December 31, 1999, followed up until age 20 years. Individuals were born without a diagnosis of intellectual disability (*International Statistical Classification of Diseases and Related Health Problems, Tenth Revision* [*ICD-10*] codes F70-F79). Analyses were conducted from May 2024 to December 2025.

EXPOSURES Age at ADHD diagnosis, identified by the first clinical diagnosis (*International Classification of Diseases, Ninth Revision* code 314; *ICD-10* code F90; or medication purchase).

MAIN OUTCOMES AND MEASURES Grade point average (GPA; range, 4 [failing] to 10 [excellent]) at the end of compulsory education (approximately 16 years of age), completed degrees in upper secondary education (vocational or academic), enrollment in tertiary education, and school dropout at age 20 years.

RESULTS The study sample included 580 132 individuals (51.2% male). Of these, 12 208 males (2.1%) and 3753 females (0.7%) had a first ADHD diagnosis between ages 4 and 20 years: mean (SD) age for males was 11.3 (4.1) years and for females was 14.4 (4.6) years. ADHD diagnosis at any age was associated with worse educational outcomes and greater likelihood of vocational than academic upper secondary education. After adjusting for sociodemographic covariates, if a diagnosis was received by age 16 years, younger age was associated with higher GPA for males (range, 7.12 [95% CI, 6.99-7.26] at age 4 years to 6.52 [95% CI, 6.46-6.58] at age 16 years) and females (range, 7.64 [95% CI, 7.49-7.78] at age 6 years to 6.95 [95% CI, 6.82-7.07]) at age 12 years, higher probability of completing an academic upper secondary degree for males (range, 20.77% [95% CI, 15.41%-26.12%] at age 4 years to 5.29% [3.78%-6.80%] at age 15 years) and females (range, 31.04% [95% CI, 15.60%-46.47%] at age 4 years to 12.01% [95% CI, 7.80%-16.21%] at age 14 years), and lower probability of school dropout for males (range, 9.16% [95% CI, 4.89%-13.42%] at age 4 years to 29.52% [95% CI, 25.85%-33.19%] at age 16 years) and females (range, 9.57% [95% CI, 4.49%-14.65%] at age 6 years to 27.16% [95% CI, 19.75%-34.57%] at age 13 years). However, after educational track choices (ages 17-20 years), older age at diagnosis was associated with higher and more academic education.

CONCLUSIONS AND RELEVANCE In this cohort study, earlier age at ADHD diagnosis was associated with better school performance, more academic education, and lower school dropout rates than diagnoses closer to age 16 years. The findings suggest that individuals who are diagnosed closer to age 16 years could benefit from targeted support to prevent school dropout.

JAMA Psychiatry. 2026;83(6):620-629. doi:10.1001/jamapsychiatry.2026.0181
Published online April 8, 2026.

Supplemental content

Author Affiliations: Helsinki Institute for Demography and Population Health, Faculty of Social Sciences, University of Helsinki, Helsinki, Finland (Volotinen, Remes, Martikainen, Metsä-Simola); Max Planck–University of Helsinki Center for Social Inequalities in Population Health, Helsinki, Finland (Volotinen, Remes, Martikainen, Metsä-Simola); Max Planck Institute for Demographic Research, Rostock, Germany (Martikainen).

Corresponding Author: Lotta Volotinen, MSc, Helsinki Institute for Demography and Population Health, Faculty of Social Sciences, University of Helsinki, PO Box 16, Helsinki, FI-00014, Finland (lotta.volotinen@helsinki.fi).

© 2026 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by SUNY Stony Brook user on 10/22/2026

Attention-deficit/hyperactivity disorder (ADHD) is a common neurodevelopmental disorder affecting individuals at all life stages.[1,2] ADHD has been associated with poorer school performance, lower attendance in higher education, and an increased likelihood of vocational than academic education.[3-11] Some studies comparing individuals who did and did not receive treatment for ADHD have suggested that educational outcomes could be improved with treatment[3-5,12]; thus, an early diagnosis is often recommended.[3-5] To our knowledge, no studies have empirically examined the association of age at first ADHD diagnosis and different educational outcomes.

ADHD is most often diagnosed during early to mid-childhood.[13-19] Receiving a diagnosis in childhood requires that symptoms, including inattention with or without hyperactive-impulsive behavior, negatively affect academic or social functioning.[1,2] Still, onset of ADHD may occur earlier than the receipt of the diagnosis; according to the current diagnostic criteria, a later diagnosis is feasible if the symptoms were present before age 12 years.[1,2] For some, a diagnosis may be delayed until adolescence due to more externally perceivable ADHD symptoms occurring only after the onset of puberty.[20] A later diagnosis may also be prompted by hindered school performance and attendance[21] that may, to some degree, be explained by the increasing demands at higher levels of education and changes in social surroundings due to school transitions. A delayed diagnosis is particularly common among females, who tend to have better school performance and fewer externally perceived symptoms than males.[13-16,20-23]

In this study, we used data from Finnish national registers on individuals born in between January 1, 1990, and December 31, 1999, as the universal education and health care systems in Finland provide an ideal setting to study the associations between age at diagnosis and educational outcomes. ADHD is commonly diagnosed in public health care, with regular screening of children's health and development and statutory health checkups with physicians around ages 4, 7, 11, and 14 years,[24] and teachers and parents are asked to pre-evaluate the student's learning and behavior (eg, participation, attendance, and taking care of belongings and assignments).[25] These questions are likely to capture some symptoms of ADHD. Education is free at all levels in Finland, with nearly all children completing compulsory basic education (around age 16 years) and most advancing to upper secondary education.[26,27] Furthermore, academic and vocational education diverge only after completing compulsory education,[26,28] avoiding the complication that early tracking might affect the likelihood of receiving a diagnosis. The current study examined (1) the incidence of ADHD diagnosis by age and (2) whether age at diagnosis was associated with grade point average (GPA) at age 16 years, vocational or academic upper secondary degree completion, tertiary education enrollment, and school dropout at age 20 years.

Key Points

Questions Is age at first attention-deficit/hyperactivity disorder (ADHD) diagnosis associated with educational outcomes?

Findings In this cohort study of 580 132 individuals, earlier age at ADHD diagnosis was associated with better grade point average at age 16 years, higher probability of academic than vocational studies, and lower probability of school dropout at age 20 years compared with diagnosis during the last years of compulsory basic education.

Meaning The findings suggest that earlier ADHD diagnosis is associated with better educational outcomes, while individuals first diagnosed toward the end of compulsory education may be at high risk of school dropout.

Methods

Data and Sample

This cohort study used data based on individual-level observations of the Finnish total population from national registers. To form the sample, we used information from the Medical Birth Register on all live-born individuals in Finland between 1990 and 1999, with follow-up until age 20 years. We identified age at first ADHD diagnosis with health care data from the Finnish Institute for Health and Welfare and ADHD medication purchases from the Social Insurance Institution of Finland. Datasets provided by Statistics Finland enabled the linkage of demographic and educational information as well as information on parental and family characteristics. Use of the pseudonymized data was approved by the ethical committees of Statistics Finland and Findata. In Finland, informed consent is not required from study participants when register data are used for scientific research carried out in public interest. This study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline.[29,30]

We excluded individuals with any diagnosis of intellectual disability (*International Statistical Classification of Diseases and Related Health Problems, Tenth Revision* (*ICD-10*) codes F70-F79), those who emigrated or died before age 20 years, and those without knowledge of their biological mother or without region of residence. We also excluded individuals who had received a diagnosis of ADHD before age 4 years, as they are more prone to misdiagnosis,[1] as well as those diagnosed after age 20 years. The final sample was followed up for their first ADHD diagnosis between ages 4 and 20 years.

Educational Outcomes

In Finland, preprimary education typically begins at age 6 years, primary education (International Standard Classification of Education [ISCED] level 1) at age 7 years, lower secondary education (ISCED level 2) at age 13 years, and upper secondary education (ISCED levels 3 and 4) at age 16 years.[28] For those born between 1990 and 1999, primary and lower secondary education were mandatory (hereafter referred to as *compulsory education*), and this compulsory schooling usually ended at age 16 years.[27] After compulsory education, individuals could apply to vocational or academic upper secondary education lasting 2 to 4 years. Tertiary education (ISCED levels ≥6), universities of applied sciences or academic universities, is usually applied for after the completion of an upper secondary degree.[26,28]

© 2026 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.
Downloaded from jamanetwork.com by SUNY Stony Brook user on 10/22/2026

GPA

School performance was measured by final grades at the end of compulsory education and was retrieved from the first applications to upper secondary education submitted from 2004 to 2019. We calculated GPA (range, 4 [failing] to 10 [excellent]) of compulsory theoretical subjects (languages, biology, mathematics, physics, chemistry, geography, history, social studies, and health education). GPA was only available for those who had applied to upper secondary education.

Completed Upper Secondary Degrees, Enrollment in Education, and School Dropout

We measured the other educational outcomes at age 20 years, when most individuals have completed upper secondary education. We identified outcomes from educational registers, including completed upper secondary degrees in vocational (no or yes) or academic education (no or yes) and current enrollment in upper secondary vocational (no or yes) or academic education (no or yes) or tertiary education (no or yes). Additionally, we measured school dropout (no or yes) as not having completed any upper secondary degrees and not being enrolled in upper secondary or tertiary education at age 20 years.

Age at ADHD Diagnosis

As in a previous study,[13] we identified ADHD diagnoses from data on public-sector specialized and primary health care and purchases of ADHD medication (data availability are given in **Figure 1** and specific registers and codes in eTable 1 in Supplement 1). Age at diagnosis refers to the calendar year of the first recorded clinical diagnosis or medication purchase.

Statistical Analysis

Analyses were conducted from May 2024 to December 2025. We used linear regression with a robust variance estimator to observe whether age at diagnosis (4-16 years for GPA and 4-20 years for other outcomes) was associated with GPA at age 16 years and logistic regression to estimate completed vocational or academic upper secondary degrees and enrollment in vocational or academic upper secondary education or tertiary education at age 20 years. Results from logistic regression models are presented as estimated probabilities that were derived using marginal standardization, which sums estimated probabilities to a weighted mean of the confounder distribution in the overall population.[31] All models included the interaction of age at diagnosis and sex (male and female) assigned at birth.

The models were adjusted for birth year (1990-1999) and region of residence (19 regions) to control for temporal changes and area-level differences in both ADHD diagnostics and educational outcomes.[26,32-34] Since the association of age at diagnosis and educational outcomes may be confounded by sociodemographic factors,[22,35,36] we further adjusted the model with the following family characteristics: maternal age at birth (<25, 25-29, 30-34, and ≥35 years), birth order (1, 2, and ≥3), number of children living in the child's household (1, 2, ≥3, and other or unknown), and living arrangements (with 2 biological parents, with 1 biological and 1 social parent, with a single parent, and other or unknown). The full model was also adjusted for the highest achieved educational level (basic, secondary, and tertiary) of biological parents and household disposable income in annual quintiles (other or unknown; 1 [lowest income] to 5 [highest income]). The number of children living in the household, living arrangements, parental educational level, and household income were measured at the beginning of the year of the individual's fourth birthday (the study design is given in Figure 1). Main findings for all educational outcomes are reported with minimum and maximum values from fully adjusted models with 95% CIs. Estimates from minimally adjusted models are shown in eTable 5 in Supplement 1 and eTables 6 to 11 in Supplement 2.

Additionally, we used a subsample of individuals aged 22 years born from 1990 to 1998 to analyze changes in completed degrees, enrollment, and school dropout at an older age, since having ADHD may hinder educational attainment but those with diagnoses may catch up with time. All analyses were conducted with Stata/MP, version 18.0 (StataCorp LLC).

Results

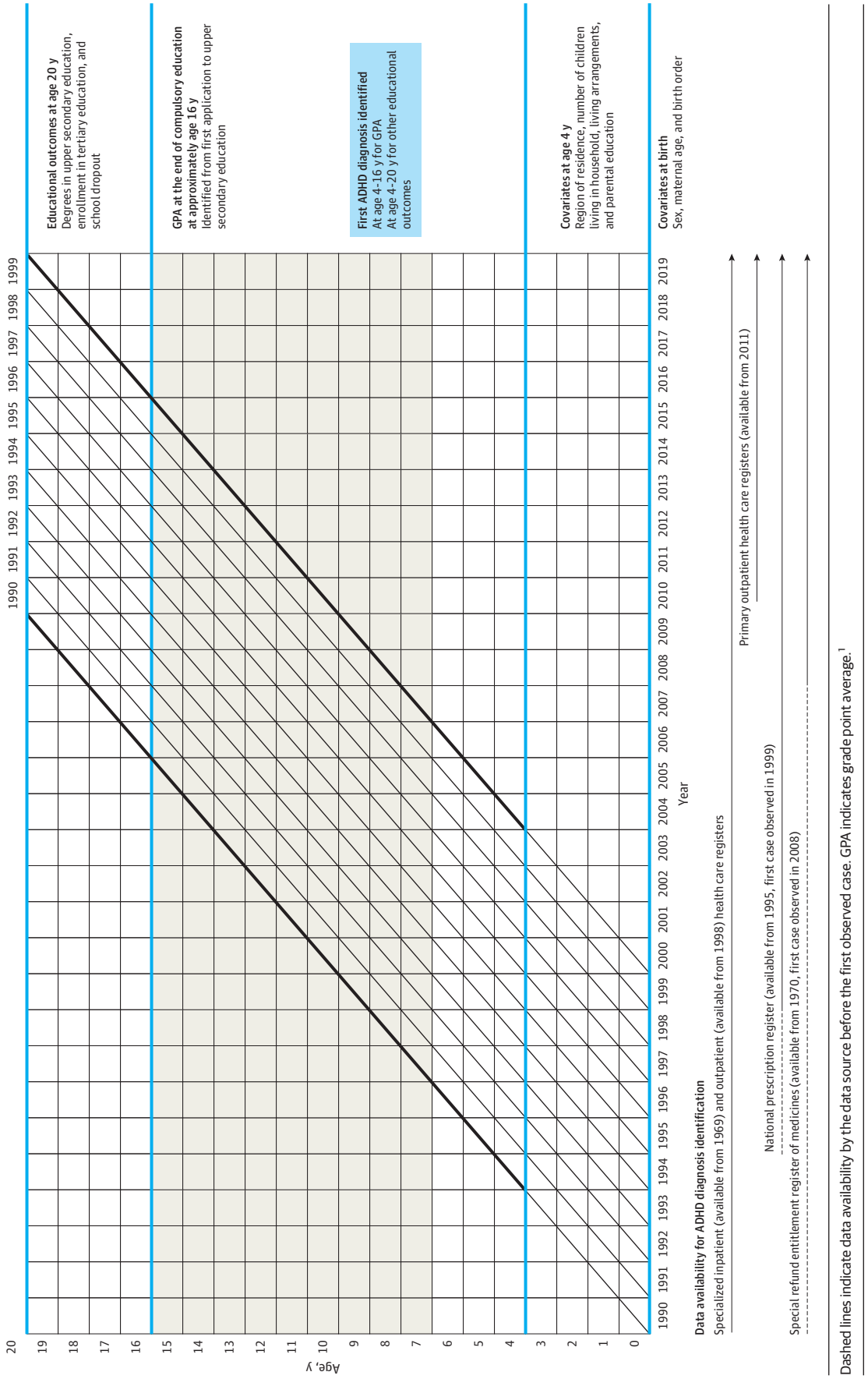
Age at ADHD Diagnosis by Sex

Among 620 656 liveborn individuals in Finland between 1990 and 1999, 6305 (1.0%) with any diagnosis of intellectual disability, 15 913 (2.6%) who emigrated, 4905 (0.7%) who died before age 20 years, 146 (0.02%) without knowledge of their biological mother, 322 (0.05%) without a region of residence, 120 (0.02%) who had received a diagnosis of ADHD before age 4 years, and 12 813 (2.1%) diagnosed after age 20 years were excluded. The final sample included 580 132 individuals followed up for their first ADHD diagnosis between ages 4 and 20 years (48.8% female; 51.2% male). Of these individuals, we identified 12 208 males (2.1%) and 3753 females (0.7%) born from 1990 to 1999 without a diagnosis of intellectual disability who were diagnosed with ADHD between ages 4 to 20 years: mean (SD) age for males was 11.3 (4.1) years and for females was 14.4 (4.6) years (**Table**). Among all 15 961 individuals, we identified the first ADHD diagnosis in 9792 from specialized health care (61.4%), 544 from primary health care (3.4%), 5620 from medication purchase (35.2%), and 5 from special reimbursement right data (0.03%) (Figure 1). Most baseline characteristics were similarly distributed by age at diagnosis (Table and eTable 2 in Supplement 2). Males were more likely to be diagnosed with ADHD during the first years of school and females, starting from the end of compulsory education (**Figure 2** and eTable 3 in Supplement 1). Among males, we observed peaks in incidence of ADHD diagnosis at age 8 years (0.44%; 95% CI, 0.41%-0.46%) and ages 14 to 15 years (0.29%; 95% CI, 0.27%-0.31%). Among females, ADHD diagnoses were less common at age 8 years (incidence, 0.07%; 95% CI, 0.06%-0.08%), but incidence increased between ages 14 years (0.07%; 95% CI, 0.06%-0.08%) and 20 years (0.16%; 95% CI, 0.15%-0.18%). The cumulative incidence of ADHD was 4.11% (95% CI, 4.04-4.18) among males and 1.33% (95% CI, 1.28-1.37) among females (Table and eTable 3 in Supplement 1).

© 2026 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by SUNY Stony Brook user on 10/22/2026

Figure 1. Lexis Diagram of the Study Design and Data Availability for Attention-Deficit/Hyperactivity Disorder (ADHD) Diagnosis Identification



Dashed lines indicate data availability by the data source before the first observed case. GPA indicates grade point average.[1]

© 2026 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.
Downloaded from jamanetwork.com by SUNY Stony Brook user on 10/22/2026

Table. Sample Characteristics by Sex and Selected Ages at ADHD Diagnosis[a]

Characteristic	Males					Females					All (N = 580 132)
	Age at ADHD diagnosis, y				No ADHD diagnosis (n = 285 038)	Age at ADHD diagnosis, y				No ADHD diagnosis (n = 279 133)	
	6 (n = 752)	11 (n = 844)	16 (n = 557)	4-20 (n = 12 208)		6 (n = 125)	11 (n = 108)	16 (n = 296)	4-20 (n = 3753)		
Incidence of ADHD, % (95% CI)	0.25 (0.24-0.27)	0.29 (0.27-0.31)	0.19 (0.18-0.21)	4.11 (4.04-4.18)[b]	NA	0.04 (0.04-0.05)	0.04 (0.03-0.05)	0.11 (0.09-0.12)	1.33 (1.28-1.37)[b]	NA	2.70 (2.66-2.75)
Birth year, mean (SD)	1995.4 (2.4)	1995.5 (2.6)	1995.1 (2.8)	1995.4 (2.8)	1994.3 (2.9)	1995.8 (2.3)	1996.0 (2.3)	1996.0 (2.7)	1995.9 (2.6)	1994.3 (2.9)	1994.4 (2.9)
Maternal age at birth, No. (%), y											
<25	206 (27.4)	249 (29.5)	166 (29.8)	3580 (29.3)	54 480 (19.1)	31 (24.8)	30 (27.8)	85 (28.7)	1038 (27.7)	53 979 (19.3)	113 077 (19.5)
25-29	258 (34.3)	275 (32.6)	200 (35.9)	4013 (32.9)	101 617 (35.7)	48 (38.4)	33 (30.6)	104 (35.1)	1224 (32.6)	99 080 (35.5)	205 934 (35.5)
30-34	190 (25.3)	218 (25.8)	124 (22.3)	3020(24.7)	85 172 (29.9)	30 (24.0)	30 (27.8)	68 (23.0)	963 (25.7)	83 158 (29.8)	172 313 (29.7)
≥35	98 (13.0)	67 (12.1)	67 (12.0)	1595 (13.1)	43 769 (15.4)	16 (12.8)	15 (13.9)	39 (13.2)	528 (14.1)	42 916 (15.4)	88 808 (15.3)
Birth order, No. (%)											
1	419 (55.7)	449 (53.2)	266 (47.8)	6453 (52.9)	138 696 (48.7)	71 (56.8)	63 (58.3)	140 (47.3)	1859 (49.5)	135 507 (48.6)	282 515 (48.7)
2	237 (31.5)	262 (31.0)	185 (33.2)	4001(32.8)	103 605 (36.4)	40 (32.0)	32 (29.6)	107 (36.2)	1255 (33.8)	101 731 (36.5)	210 605 (36.3)
≥3	96 (12.8)	133 (15.8)	106 (19.0)	1754(14.4)	42 737 (15.0)	14 (11.2)	13 (12.0)	49 (16.6)	623 (16.7)	41 895 (15.0)	87 012 (15.0)
No. of children in household (%)											
1	201 (26.7)	220 (26.1)	127 (22.8)	3034(25.5)	49 857 (17.5)	33 (26.4)	30 (27.8)	64 (21.6)	571 (24.7)	49 371 (17.7)	103 271 (17.8)
2	353 (46.9)	361 (42.8)	249 (44.7)	5518 (45.2)	137 483 (48.2)	53 (42.4)	52 (48.2)	147 (49.7)	1691 (45.1)	133 488 (47.8)	278 080 (47.9)
≥3	183 (24.3)	245 (29.0)	169 (30.3)	3345 (27.4)	96 296 (33.8)	33 (26.4)	NR	80 (27.0)	978 (28.3)	94 905 (34.0)	195 607 (33.7)
Other or unknown	15 (2.0)	18 (2.1)	12 (2.2)	224 (1.9)	1402 (0.5)	6 (4.8)	NR	5 (1.7)	58 (2.0)	1469 (0.5)	3174 (0.6)
Living arrangements, No. (%)											
2 Biological parents	536 (71.3)	573 (67.9)	389 (69.8)	8678(71.1)	247 806 (86.9)	82 (65.6)	70 (64.8)	204 (68.9)	2664 (71.0)	241 299 (86.5)	500 447 (86.3)
1 Biological and 1 social parent	22 (2.9)	34 (4.0)	19 (3.4)	413 (3.5)	4231 (1.5)	5 (4.0)	NR	12 (4.1)	77 (3.5)	4445 (1.6)	9230 (1.6)
1 Biological parent	177 (23.5)	218 (25.8)	136 (24.4)	2852 (23.4)	31 313 (11.0)	31 (24.8)	29 (26.9)	75 (25.3)	859 (23.2)	31 628 (11.3)	66 662 (11.5)
Other or unknown	17 (2.3)	19 (2.3)	13 (2.3)	254 (2.1)	1688 (0.6)	7 (5.6)	NR	5 (1.7)	52 (2.3)	1761 (0.6)	3793 (0.7)
Highest parental educational level, No. (%)											
Basic	109 (14.5)	119 (14.1)	75 (13.5)	1609 (13.2)	18 964 (6.7)	20 (16.0)	24 (22.2)	30 (10.1)	483 (13.0)	19 003 (6.8)	40 062 (6.9)
Secondary	523 (69.6)	588 (69.7)	384 (68.9)	8559 (70.1)	197 271 (69.2)	95 (76.0)	67 (62.0)	199 (67.2)	2597 (69.2)	194 006 (69.5)	402 433 (69.4)
Tertiary	120 (16.0)	137 (16.2)	98 (17.6)	2040 (16.7)	68 803 (24.1)	10 (8.0)	17 (15.7)	67 (22.6)	666 (17.9)	66 124(23.7)	137 637 (23.7)
Household income quintile, No. (%)											
First (lowest)	163 (21.7)	235 (27.8)	147 (26.4)	2972 (12.1)	48 700 (17.1)	30 (24.0)	32 (29.6)	88 (29.7)	930 (9.1)	48 967 (17.5)	101 569 (17.5)
Second	206 (27.4)	213 (25.2)	137 (24.6)	3223(23.1)	67 964 (23.8)	31 (24.8)	27 (25.0)	68 (23.0)	978 (23.8)	66 484 (23.8)	138 649 (23.9)
Third	165 (21.9)	176 (20.9)	124 (22.3)	2609 (23.9)	66 515 (23.3)	29 (23.2)	16 (14.8)	59 (19.9)	796 (25.6)	65 222 (23.4)	137 637 (23.3)
Fourth	134 (17.8)	129 (15.3)	75 (13.5)	2021 (18.7)	59 204 (20.8)	23 (18.4)	16 (14.8)	48 (16.2)	622 (18.7)	57 424 (20.6)	119 271 (20.6)
Fifth (highest)	72 (9.6)	84 (10.0)	61 (11.0)	1126 (13.9)	41 533 (14.6)	6 (4.8)	NR	NR	80 (14.5)	39 846 (14.3)	82 920 (14.3)
Other or unknown	12 (1.6)	7 (0.8)	13 (2.3)	194 (8.3)	1122 (0.4)	6 (4.8)	NR	NR	31 (8.4)	1190 (0.4)	2577 (0.4)

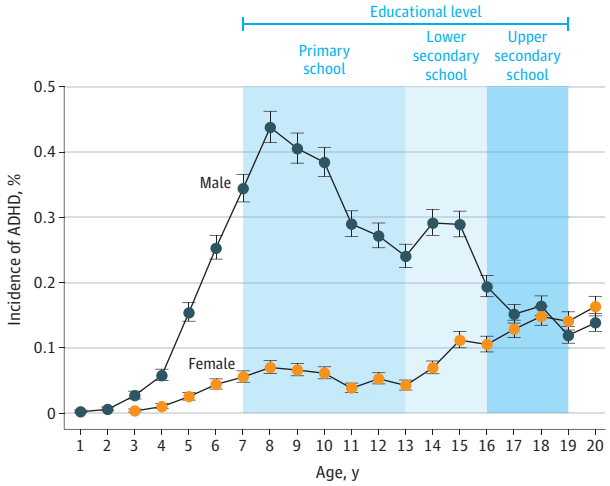
Abbreviations: ADHD, attention-deficit/hyperactivity disorder; NA, not applicable; NR, not reported (censored due to data protection).

[a] Data are reported as number (percentage) of individuals unless otherwise indicated. Data for all ages are listed in eTables 2 and 3 in Supplements 1 and 2, respectively. Distributions for birth year and region of residence are not reported due to individual's data protection.

[b] Cumulative incidence of ADHD at ages 4 to 20 years in sex groups (eTable 3 in Supplement 1).

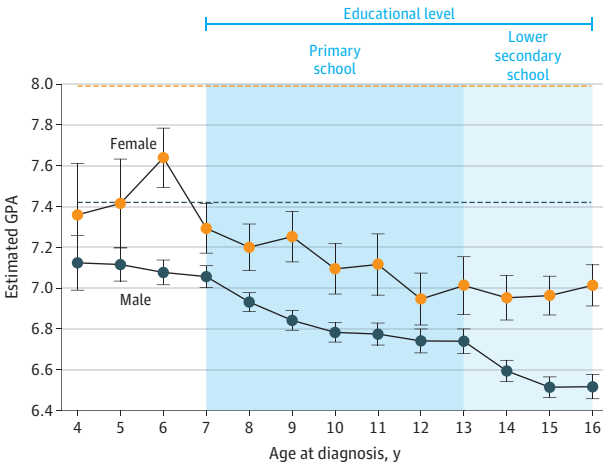
© 2026 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.
Downloaded from jamanetwork.com by SUNY Stony Brook user on 10/22/2026

Figure 2. Line Graph Showing Incidence of Attention-Deficit/Hyperactivity Disorder (ADHD) Diagnosis by Age and Sex in Birth Cohorts from 1990 to 1999 in Finland



Values with fewer than 5 observations were censored. Incidence rate values are listed in eTable 3 in Supplement 1. Error bars represent 95% CIs.

Figure 3. Line Graph Showing Linear Estimates of Grade Point Average (GPA) at Age 16 Years by Age at Attention-Deficit/Hyperactivity Disorder (ADHD) Diagnosis and Sex



Models were adjusted for birth year, region of residence, maternal age at birth, birth order, number of children in household, living arrangements, parental educational level, and household income. Model estimates are listed in eTable 5 in Supplement 1. Dashed lines represent the reference groups of individuals without a diagnosis of ADHD. Error bars represent 95% CIs.

GPA at the End of Compulsory Education

GPA was only available for those who had applied to upper secondary education, representing 98.7% of the sample. Having an ADHD diagnosis was associated with a lower GPA compared with having no diagnosis among both males and females (eTable 4 in Supplement 1). Older age at diagnosis was associated with a decreasing GPA between ages 4 years (7.12; 95% CI, 6.99-7.26) and 16 years (6.52; 95% CI, 6.46-6.58) for males and between ages 6 years (7.64, 95% CI, 7.49-7.78) and 12 years (6.95; 95% CI, 6.82-7.07) for females (**Figure 3** and eTable 5 in Supplement 1).

Degrees in Vocational and Academic Upper Secondary Education

At age 20 years, males and females diagnosed with ADHD were more likely to have a degree in vocational upper secondary education than in academic upper secondary education (eTable 4 in Supplement 1). Diagnosis at any age was associated with having a lower probability of a degree in academic upper secondary education compared with no diagnosis in males and females, but probabilities of a vocational degree were similar (eTables 6 and 7 in Supplement 2). Older age at diagnosis was associated with a decreasing probability of an academic upper secondary degree for males (range, 20.77% [95% CI, 15.41%-26.12%] at age 4 years to 5.29% [95% CI, 3.78%-6.80%] at age 15 years) and females (range, 31.04% [95% CI, 15.60%-46.47%] at age 4 years to 12.01% [95% CI, 7.80%-16.21%] at age 14 years) except among those diagnosed after the completion of compulsory education (Figure 3 and eTables 6 and 7 in Supplement 2). Upper secondary education is usually completed by age 20 years, but among males and females with an ADHD diagnosis, enrollment in vocational upper secondary education at age 20 years was common (eTable 4 in Supplement 1). Earlier age at diagnosis was associated with a higher probability of enrollment in vocational education for females, while there was no association among males (eTable 8 in Supplement 2). The probability of enrollment in academic secondary education was low but slightly elevated among those diagnosed after compulsory education (eTable 9 in Supplement 2).

Enrollment in Tertiary Education

Continuation to tertiary education was less likely among those with an ADHD diagnosis than among peers with no ADHD diagnosis (eTable 4 in Supplement 1). For both males and females diagnosed at any age, the probability of enrollment in tertiary education remained below 11%, with the lowest probabilities observed for males diagnosed around the end of compulsory education at ages 14 to 17 years at approximately 3% (**Figure 4** and eTable 10 in Supplement 2).

School Dropout

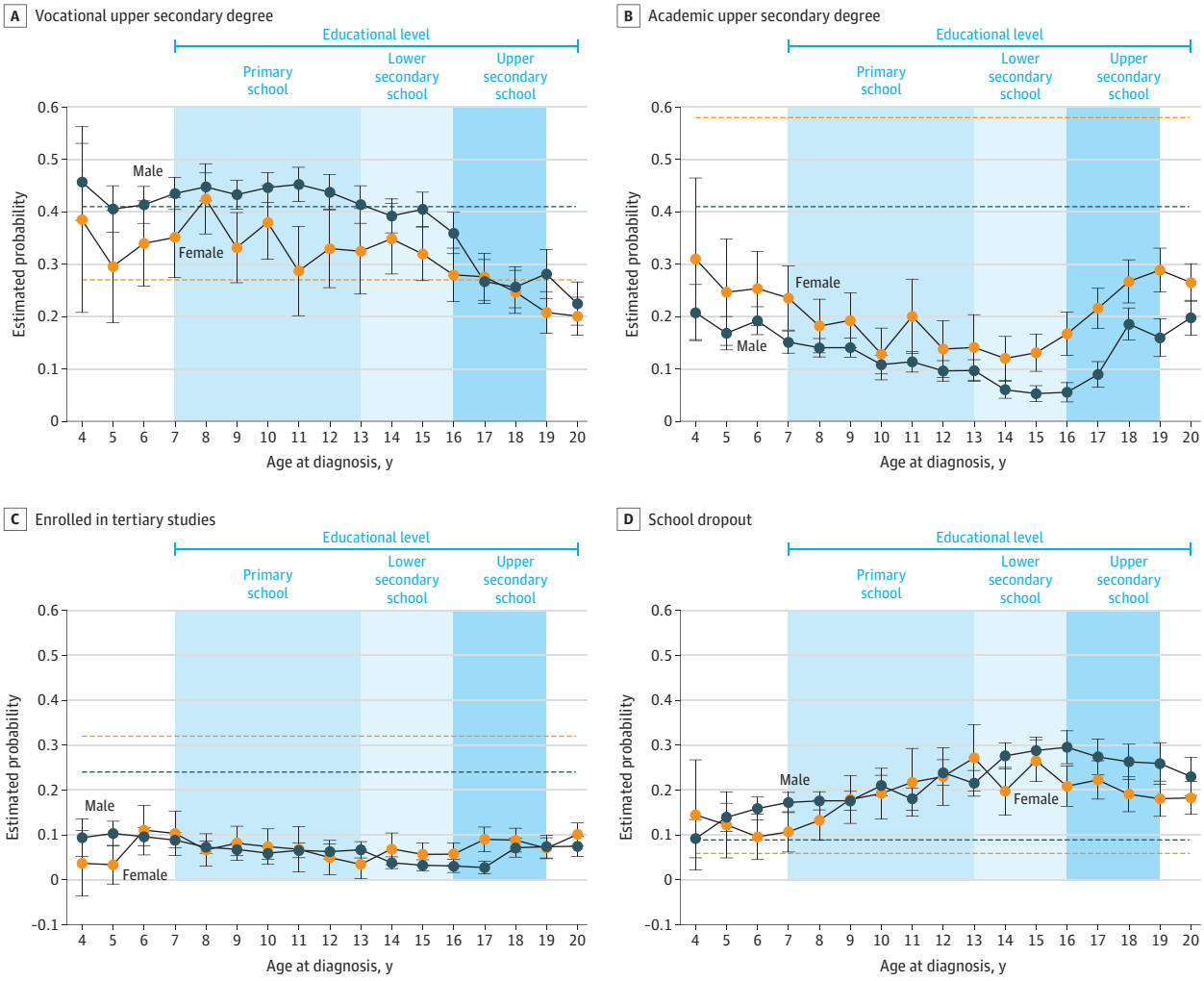
At age 20 years, having no upper secondary degree and no enrollment in education was more likely for males and females with an ADHD diagnosis compared with their peers with no ADHD diagnosis (eTable 4 in Supplement 1). Older age at diagnosis was associated with an increased probability of school dropout among males diagnosed between ages 4 years (9.16%; 95% CI, 4.89%-13.42%) and 16 years (29.52%; 95% CI, 25.85%-33.19%) and among females between ages 6 years (9.57%; 95% CI, 4.49%-14.65%) and 13 years (27.16%; 95% CI, 19.75%-34.57%) (Figure 4 and eTable 11 in Supplement 2). Among those diagnosed after compulsory education, older age at diagnosis was associated with decreasing probability of school dropout.

Additional Analyses

The results were similar in models without adjustment for family characteristics and socioeconomic position (models 1 and

© 2026 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.
Downloaded from jamanetwork.com by SUNY Stony Brook user on 10/22/2026

Figure 4. Line Graphs Showing Estimated Probabilities of Educational Outcomes at Age 20 Years by Age at Attention-Deficit/Hyperactivity Disorder (ADHD) Diagnosis and Sex



Models were adjusted for birth year, region of residence, maternal age at birth, birth order, number of children in household, living arrangements, parental education, and household income. Model estimates are listed in eTables 6, 7, 10, and 11 in Supplement 2. Dashed lines represent the reference group of individuals without a diagnosis of ADHD. Error bars represent 95% CIs.

2 in eTable 5 in Supplement 1 and eTables 6-11 in Supplement 2). When completed degrees, educational enrollment, and school dropout were measured at age 22 years for cohorts born between 1990 and 1998, an additional 10.00% of males and 12.65% of females with an ADHD diagnosis had completed an upper secondary degree; however, overall, the pattern of results remained similar (eTables 12-17 in Supplement 1).

Discussion

To our knowledge, our study is the first to report the associations of age at ADHD diagnosis and educational outcomes. The results showed that although individuals with an ADHD diagnosis had poorer educational outcomes and chose academic studies less often than those without a diagnosis,[3-11] the outcomes differed by the age at diagnosis. Individuals diagnosed with ADHD at the beginning of compulsory education had better school performance, were more likely to continue their studies in upper secondary or tertiary education, and were less likely to drop out of school than those diagnosed toward the end of compulsory education. These associations were independent of covariates relating to family characteristics and socioeconomic position. The findings support the idea that an early diagnosis could be beneficial for later educational outcomes.[3-5]

An earlier diagnosis may be associated with improved educational outcomes through multiple pathways. First, an ADHD diagnosis is usually required to access pharmacologic and nonpharmacologic treatment, which may improve educational outcomes of the individuals with an ADHD diagnosis; however, more studies are needed on the longer-lasting effects.[3-5,12] Second, support measures in schools may become

© 2026 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by SUNY Stony Brook user on 10/22/2026

more easily available with a diagnosis. Third, an earlier diagnosis may lead to greater awareness of ADHD for the individuals and their families, which may have positive long-term effects on educational outcomes through improving the individual's self-awareness, self-acceptance, and self-efficacy as well as family and other relationship dynamics.[37-41]

In our study, the poorest educational outcomes were observed among those who were diagnosed with ADHD toward the end of compulsory education. At age 20 years, up to one-third (29.52% at age 16 years among males and 27.16% at age 13 years among females) had neither completed an upper secondary degree nor were enrolled in education, which may increase the risk of unemployment and long-term marginalization. Poorer school performance at the end of compulsory education has been observed to mediate the association between having an ADHD diagnosis and educational outcomes.[11] Those who were diagnosed toward the end of compulsory education had the least amount of time to benefit from the diagnosis and improve their school performance before the end of compulsory education. These individuals may need additional preparatory education before applying to upper secondary education[28] as well as additional support to complete upper secondary education.

After the end of compulsory education and applications to upper secondary education, older age at diagnosis was associated with a greater probability of enrollment in academic upper secondary and tertiary education and lower probability of school dropout. This finding may be due to detection bias, whereby students who perform well in lower grades receive late diagnoses when faced with more academically demanding settings at higher education levels. However, those who are diagnosed with ADHD after compulsory education may still require more time to complete their studies: by age 22 years, an additional 10.00% of males and 12.65% of females diagnosed with ADHD had finished an upper secondary degree, whereas most of their peers without an ADHD diagnosis had already finished a degree by age 20 years.

In accordance with earlier evidence,[6,7] we found that on the absolute scale, males with an ADHD diagnosis performed worse in school than females with a diagnosis, but on the relative scale, females with an ADHD diagnosis performed worse than females without a diagnosis. Still, the association between age at diagnosis and school performance was similar for males and females. Similarly, as in earlier studies,[13-16] males were also diagnosed with ADHD most often during the first years of school and females increasingly in adolescence after age 13 years. Although a later diagnosis may be prompted by more externally perceived ADHD symptoms in puberty,[20,23] changes in social surroundings due to school transitions, or increasing demands at higher levels of education, we observed increases in diagnoses around and after the statutory health checkups (ages 4, 7, 11, and 14 years), suggesting that screening for ADHD before age 12 years could be beneficial. Future studies should try to identify family-related barriers to seeking or accessing care and the aspects of service provision that may delay a diagnosis.

Strengths and Limitations

A major strength of our study is the individual-level, register-based data from Finland, which covers the total population and provides detailed information on health and education without self-report bias. With these data, we could study 10 cohorts of individuals for 20 years with minimal loss to follow-up.

This study also has limitations. GPA was missing for individuals who did not apply through the Finnish joint application system by age 20 years (1.3% of our sample); this likely made our GPA results more conservative, since ADHD was a risk factor for not applying. However, the proportion not applying did not systematically vary by age at diagnosis (eTable 18 in Supplement 1), and thus, lack of data for these individuals was unlikely to have biased the observed age pattern.

Furthermore, our data did not include clinical diagnoses made in primary health care before 2011 or in private clinics. However, the diagnostics of ADHD shifted from specialized to primary health care, mainly after the 2010s.[32,42] Moreover, our data included complete information on medication purchases, and thus, we only missed individuals who were diagnosed in private clinics and had no medication purchases. We believe that under-ascertainment was likely to have been limited, as the cumulative incidence of ADHD was 4.11% among males and 1.33% among females, corresponding with previously reported national prevalence estimates.[33] Still, for some individuals, the observed age at diagnosis may have been later than the true age. This time gap can be shorter due to more perceived symptoms that are associated with medication use.[43] If age at diagnosis was systematically lower among individuals with more symptoms due to their higher probability of beginning medication use, the estimates showing that educational outcomes were better among those with an earlier diagnosis are likely to be conservative. Finally, our study design did not allow us to assess causality of the observed associations, and more research is needed with methods better suited for causal inference.

Conclusions

In this cohort study based on total population register data, we observed that age at first ADHD diagnosis was associated with educational outcomes among young individuals in Finland. ADHD diagnosis during the first years of school was associated with better school performance, more academic track choices, and lower probability of school dropout. The findings support the recommendations for earlier diagnosis, and screening for ADHD before age 12 years should be considered. Additional support should be directed to individuals who are diagnosed during the last years of compulsory education, as they were most at risk of school dropout in our study.

ARTICLE INFORMATION

Accepted for Publication: January 1, 2026.

Published Online: April 8, 2026. doi:10.1001/jamapsychiatry.2026.0181

Author Contributions: Ms Volotinen had full access to all of the data in the study and takes

© 2026 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by SUNY Stony Brook user on 10/22/2026

responsibility for the integrity of the data and the accuracy of the data analysis.
Concept and design: All authors.
Acquisition, analysis, or interpretation of data: All authors.
Drafting of the manuscript: Volotinen.
Critical review of the manuscript for important intellectual content: All authors.
Statistical analysis: Volotinen.
Obtained funding: Martikainen, Metsä-Simola.
Supervision: Remes, Martikainen, Metsä-Simola.

Conflict of Interest Disclosures: Dr Martikainen reported receiving grants from the European Research Council, Strategic Research Council within the Research Council of Finland, Research Council of Finland, Jane and Aatos Erkko Foundation, Max Planck Society, University of Helsinki, and Cities of Helsinki, Vantaa, and Espoo during the conduct of the study. Dr Metsä-Simola reported receiving grants from Research Council of Finland during the conduct of the study. No other disclosures were reported.

Funding/Support: Ms Volotinen was funded by the Doctoral Programme in Social Sciences, University of Helsinki, and gratefully acknowledges the resources provided by the International Max Planck Research School for Population, Health and Data Science (IMPRS-PHDS). Drs Martikainen and Remes were supported by grant 101019329 from the European Research Council under the European Union's Horizon 2020 research and innovation programme, and grants 210046 to the Max Planck–University of Helsinki Center from the Jane and Aatos Erkko Foundation, 77204227 from the Max Planck Society, and 5714240218 from the University of Helsinki, and Cities of Helsinki, Vantaa, and Espoo. Dr Martikainen was also supported by grants 352543-352572 (for ACElife) and 345219 (for LIFECON) from the Strategic Research Council within the Academy of Finland. Dr Metsä-Simola was supported by grant 363724 from the Research Council of Finland.

Role of the Funder/Sponsor: The funders had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Disclaimer: The study does not necessarily reflect the European Commission's views and in no way anticipates the European Commission's future policy in this area.

Data Sharing Statement: See Supplement 3.

REFERENCES

1. American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5TM)*. 5th ed. American Psychiatric Association; 2013.

2. World Health Organization. *International Classification of Diseases, Eleventh Revision (ICD-11) for Morbidity and Mortality Statistics*. 2019. Accessed May 15, 2025. https://icd.who.int/browse/2025-01/mms/en

3. Arnold LE, Hodgkins P, Kahle J, Madhoo M, Kewley G. Long-term outcomes of ADHD: academic achievement and performance. *J Atten Disord*. 2020;24(1):73-85. doi:10.1177/1087054714566076

4. Jangmo A, Stålhandske A, Chang Z, et al. Attention-deficit/hyperactivity disorder, school performance, and effect of medication. *J Am Acad Child Adolesc Psychiatry*. 2019;58(4):423-432. doi:10.1016/j.jaac.2018.11.014

5. Shaw M, Hodgkins P, Caci H, et al. A systematic review and analysis of long-term outcomes in attention deficit hyperactivity disorder: effects of treatment and non-treatment. *BMC Med*. 2012;10(1):99-99. doi:10.1186/1741-7015-10-99

6. Sunde HF, Kleppestø TH, Gustavson K, Nordmo M, Reme BA, Torvik FA. The ADHD deficit in school performance across sex and parental education: a prospective sibling-comparison register study of 344,152 Norwegian adolescents. *JCPP Adv*. 2022;2(1):e12064. doi:10.1002/jcv2.12064

7. Nordmo M, Kleppestø TH, Reme BA, Sunde HF, von Soest T, Torvik FA. The diminishing association between adolescent mental disorders and educational performance from 2006-2019. *JCPP Adv*. 2024;4(3):e12239. doi:10.1002/jcv2.12239

8. Fleming M, Fitton CA, Steiner MFC, et al. Educational and health outcomes of children treated for attention-deficit/hyperactivity disorder. *JAMA Pediatr*. 2017;171(7):e170691. doi:10.1001/jamapediatrics.2017.0691

9. Mikkonen J, Moustgaard H, Remes H, Martikainen P. The population impact of childhood health conditions on dropout from upper-secondary education. *J Pediatr*. 2018;196:283-290.e4. doi:10.1016/j.jpeds.2018.01.034

10. Ringbom I, Suvisaari J, Kääriälä A, et al. Psychiatric disorders diagnosed in adolescence and subsequent long-term exclusion from education, employment or training: longitudinal national birth cohort study. *Br J Psychiatry*. 2022;220(3):148-153. doi:10.1192/bjp.2021.146

11. Mikkonen J, Remes H, Moustgaard H, Martikainen P. Early adolescent health problems, school performance, and upper secondary educational pathways: a counterfactual-based mediation analysis. *Soc Forces*. 2021;99(3):1146-1175.

12. Langberg JM, Becker SP. Does long-term medication use improve the academic outcomes of youth with attention-deficit/hyperactivity disorder? *Clin Child Fam Psychol Rev*. 2012;15(3):215-233. doi:10.1007/s10567-012-0117-8

13. Volotinen L, Remes H, Martikainen P, Metsä-Simola N. Parental education level and ADHD diagnosis in childhood and adolescence: the moderating roles of gender, age, and family history of ADHD. *Eur Child Adolesc Psychiatry*. Published online September 18, 2025. doi:10.1007/s00787-025-02852-0

14. Dalsgaard S, Thorsteinsson E, Trabjerg BB, et al. Incidence rates and cumulative incidences of the full spectrum of diagnosed mental disorders in childhood and adolescence. *JAMA Psychiatry*. 2020;77(2):155-164. doi:10.1001/jamapsychiatry.2019.3523

15. Skoglund C, Sundström Poromaa I, Leksell D, et al. Time after time: failure to identify and support females with ADHD—a Swedish population register study. *J Child Psychol Psychiatry*. 2024;65(6):832-844. doi:10.1111/jcpp.13920

16. Westman E, Prami T, Kallio A, et al. Increase in occurrence of attention deficit hyperactivity disorder differs by age group and gender-Finnish Nationwide Register Study. *Brain Behav*. 2025;15(1):e70253. doi:10.1002/brb3.70253

17. Lindemann C, Langner I, Kraut AA, et al. Age-specific prevalence, incidence of new diagnoses, and drug treatment of attention-deficit/hyperactivity disorder in Germany. *J Child Adolesc Psychopharmacol*. 2012;22(4):307-314. doi:10.1089/cap.2011.0064

18. McKechnie DGJ, O'Nions E, Dunsmuir S, Petersen I. Attention-deficit hyperactivity disorder diagnoses and prescriptions in UK primary care, 2000-2018: population-based cohort study. *BJPsych Open*. 2023;9(4):e121. doi:10.1192/bjo.2023.512

19. Rocco I, Corso B, Bonati M, Minicuci N. Time of onset and/or diagnosis of ADHD in European children: a systematic review. *BMC Psychiatry*. 2021;21(1):575. doi:10.1186/s12888-021-03547-x

20. Murray AL, Booth T, Eisner M, Auyeung B, Murray G, Ribeaud D. Sex differences in ADHD trajectories across childhood and adolescence. *Dev Sci*. 2019;22(1):e12721. doi:10.1111/desc.12721

21. Rucklidge JJ. Gender differences in attention-deficit/hyperactivity disorder. *Psychiatr Clin North Am*. 2010;33(2):357-373. doi:10.1016/j.psc.2010.01.006

22. Tam LYC, Taechameekietichai Y, Allen JL. Individual child factors affecting the diagnosis of attention deficit hyperactivity disorder (ADHD) in children and adolescents: a systematic review. *Eur Child Adolesc Psychiatry*. 2025;34(5):1469-1496. doi:10.1007/s00787-024-02590-9

23. Millenet S, Laucht M, Hohm E, et al. Sex-specific trajectories of ADHD symptoms from adolescence to young adulthood. *Eur Child Adolesc Psychiatry*. 2018;27(8):1067-1075. doi:10.1007/s00787-018-1129-9

24. Finlex. Statute Book of Finland 338/2011. 2011. Accessed April 16, 2025. https://www.finlex.fi/fi/lainsaadanto/saadoskokoelma/2011/338

25. Hakulinen-Viitanen T, Hietanen-Peltola M, Hastrup A, Wallin M, Pelkonen M. Comprehensive health check-up—instructions for maternity and child health clinics and school health care. 2012. Accessed November 30, 2025. https://www.julkari.fi/handle/10024/90831

26. OECD. Education policy outlook in Finland. 2020. Accessed April 6, 2025. https://www.oecd.org/en/publications/education-policy-outlook-in-finland_f162c72b-en.html

27. Finlex. Basic Education Act 628/1998. 1998. Accessed April 15, 2025. https://www.finlex.fi/fi/lainsaadanto/saadoskokoelma/1998/628

28. Finnish National Agency for Education. Finnish education system. Accessed May 16, 2025. https://www.oph.fi/en/education-system

29. von Elm E, Altman DG, Egger M, Pocock SJ, Gøtzsche PC, Vandenbroucke JP; STROBE Initiative. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. *Lancet*. 2007;370(9596):1453-1457. doi:10.1016/S0140-6736(07)61602-X

30. Vandenbroucke JP, von Elm E, Altman DG, et al; STROBE Initiative. Strengthening the Reporting of Observational Studies in Epidemiology (STROBE): explanation and elaboration. *Epidemiology*. 2007;18(6):805-835. doi:10.1097/EDE.0b013e3181577511

31. Muller CJ, MacLehose RF. Estimating predicted probabilities from logistic regression: different

© 2026 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by SUNY Stony Brook user on 10/22/2026

methods correspond to different target populations. *Int J Epidemiol*. 2014;43(3):962-970. doi:10.1093/ije/dyu029

32. The Finnish Medical Society Duodecim. Attention-deficit hyperactivity disorder: current care guidelines. 2025. Accessed November 30, 2025. http://www.kaypahoito.fi

33. National Institute for Health and Welfare THL. Prevalence of ADHD among children and adolescents in 2022: ADHD diagnoses are increasing steadily; gender and regional differences are quite large. Julkari. 2024. Accessed November 30, 2025. https://urn.fi/URN:NBN:fi-fe202401183216

34. Hakulinen T, Hietanen-Peltola M, Vaara S, Merikukka M, Pelkonen M; THL. Modern practices and long traditions in health clinics and school health care: service monitoring study 2016-2017. Julkari. 2018. Accessed November 24, 2025. https://www.julkari.fi/handle/10024/137282

35. Claussen AH, Holbrook JR, Hutchins HJ, et al. All in the family? A systematic review and meta-analysis of parenting and family environment as risk factors for attention-deficit/hyperactivity disorder (ADHD) in children. *Prev Sci*. 2024;25(suppl 2):249-271. doi:10.1007/s11121-022-01358-4

36. Esch P, Bocquet V, Pull C, et al. The downward spiral of mental disorders and educational attainment: a systematic review on early school leaving. *BMC Psychiatry*. 2014;14(1):237-237. doi:10.1186/s12888-014-0237-4

37. Hamed AM, Kauer AJ, Stevens HE. Why the diagnosis of attention deficit hyperactivity disorder matters. *Front Psychiatry*. 2015;6:168. doi:10.3389/fpsyt.2015.00168

38. Fleischmann A, Fleischmann RH. Advantages of an ADHD diagnosis in adulthood: evidence from online narratives. *Qual Health Res*. 2012;22(11):1486-1496. doi:10.1177/1049732312457468

39. Dangmann CR, Skogli GKW, Holthe MEG, Steffenak AKM, Andersen PN. Life gets better: important resilience factors when growing up with ADHD. *J Atten Disord*. 2024;28(8):1198-1209. doi:10.1177/10870547241246645

40. Grønneberg SV, Engebretsen E, Løkkeberg ST. Stories of hope: young people's personal narratives about ADHD put into context of positive aspects. *Qual Health Res*. 2024;34(1-2):48-60. doi:10.1177/10497323231206936

41. Ringer N. Living with ADHD: a meta-synthesis review of qualitative research on children's experiences and understanding of their ADHD. *Int J Disabil Dev Educ*. 2020;67(2):208-224. doi:10.1080/1034912X.2019.1596226

42. Auro K, Holopainen I, Perola M, Havulinna AS, Raevuori A. Attention-deficit/hyperactivity disorder diagnoses in Finland during the COVID-19 pandemic. *JAMA Netw Open*. 2024;7(6):e2418204. doi:10.1001/jamanetworkopen.2024.18204

43. Efron D, Gulenc A, Sciberras E, et al. Prevalence and predictors of medication use in children with attention-deficit/hyperactivity disorder: evidence from a community-based longitudinal study. *J Child Adolesc Psychopharmacol*. 2019;29(1):50-57. doi:10.1089/cap.2018.0095

© 2026 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by SUNY Stony Brook user on 10/22/2026